PYTHON DATA ANALYSIS PROJECT

Food Recall Root-Cause Intelligence

Exploratory Data Analysis of FDA Food Enforcement Reports

INDUSTRY

Food Manufacturing & Food Safety

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Industry Overview & Problem Statement

SELECTED INDUSTRY

Food Manufacturing & Food Safety

Food companies must keep products safe, correctly labelled and traceable. When they fail, the product is recalled: pulled from the market at the cost of lost stock, regulatory exposure and brand trust.

WHY DATA ANALYSIS MATTERS

- Shows which causes and products keep recurring
- Turns reactive recalls into preventive controls
- Supports QA teams such as Nestlé Global Food Safety & QA

What is the problem?

Raw FDA recall records are typed in by hand: free-text reasons, dates stored as text, state abbreviations and many blank or "Unknown" fields hide which causes and products really drive recalls.

Why does it matter?

Each recall means lost product, regulatory exposure and brand damage, and charts built on inconsistent data can point teams at the wrong risks.

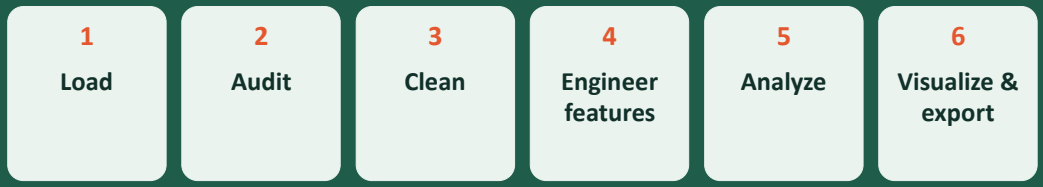
Who is affected?

Food manufacturers and their QA teams (this study is framed for Nestlé), the FDA that classifies recalls, and consumers exposed to unsafe products.

Objectives, Solution & Analysis Questions

Proposed solution

A Python pipeline in Jupyter (Pandas, NumPy, Matplotlib) that audits the raw FDA data, cleans it, derives root cause and product category from free text, and turns it into charts.



Project objectives

1. Load and understand the raw FDA data: structure, types, size
2. Audit quality: missing values, duplicates, inconsistent labels and dates
3. Clean and standardize dates, text, state names and gaps
4. Engineer features: root cause, product category, classification days
5. Explore and visualize patterns to guide preventive decisions

Analysis questions

- **Q1** Which root causes lead to the most recalls?
- **Q2** Which product categories are recalled most?
- **Q3** Which firms and states have the highest volume?
- **Q4** How do recall status and classification differ?
- **Q5** How long does the FDA take to classify a recall?
- **Q6** How has recall volume changed over the years?

Dataset & Data Understanding

Dataset	FDA Food Enforcement Reports (Food Recalls)
Source	U.S. FDA via openFDA (food/enforcement), public data
File format	CSV, converted from JSON with values unchanged
Size	29,471 rows × 24 columns (29 after feature engineering)
Grain	One row per recalled product; 7,898 recall events
Period	Mostly 2012-2026 (plus 116 records from 2008-2011)
Key columns	recall_number, event_id, status, classification, recalling_firm, product_description, product_quantity, reason_for_recall, state, four date columns
How collected	Downloaded from the openFDA endpoint, saved as CSV and loaded with pd.read_csv(dtype=str) so dates like 20240115 stay intact

0
duplicate rows found

24/24
columns loaded as text

6.0%
quantity blank, +1,492 "unknown"

5.1%
no termination date (still open)

```
In [11]: print("Number of Rows:", data.shape[0])
         print("Number of Columns:", data.shape[1])

Number of Rows: 29471
Number of Columns: 24
```

```
In [12]: # missing_summary (only columns with gaps)
         missing_summary[missing_summary["Missing Values"] > 0]
```

	Missing Values	Missing %
state	426	1.45
address_2	27704	94.00
postal_code	445	1.51
product_quantity	1781	6.04
code_info	401	1.36
more_code_info	29470	100.00
termination_date	1513	5.13

```
In [13]: print("Number of duplicate rows:", data.duplicated().sum())
         print("Unique Recall Numbers:", data["recall_number"].nunique())

Number of duplicate rows: 0
Unique Recall Numbers: 29469
```

Jupyter output: dimensions, columns with missing values, duplicate check

Quality notes: address_2 is 94% blank, state 1.45%, termination_date blank for open recalls, and one date (year 212) is implausible.

Data Cleaning & Transformation

1 Working copy

raw_data kept untouched; all edits made on clean_data (29,471 rows)

2 Column names

Stripped, lower-cased, spaces and hyphens turned into underscores

3 Data types

4 date columns text → datetime (YYYYMMDD); 0 invalid values

4 Inconsistent values

Spaces trimmed in 10 text columns; state codes → full names (CA → California)

5 Missing values

Text gaps labelled "Unknown" / "Not Reported"; dates and class left empty, not guessed

6 Duplicates & filtering

0 duplicate rows, no records dropped; negative day gaps set to NaN (0 found)

7 New columns

classification_days, recall_year, root_cause (10 groups), product_category (11), timely flag (30 days)

8 Validate & export

Rows, types and derived features checked; saved as Food_Recalls_Cleaned.csv (29 columns)

```
In [42]: clean_data["root_cause"] = "Other / Unclassified"
         clean_data.loc[reason.str.contains("label|mislabel|...
         # ... 9 more ordered keyword rules ...
         clean_data["root_cause"].value_counts()
```

root_cause	
Undeclared Allergen	9531
Listeria	7397
Other / Unclassified	3211
Salmonella	2969
Foreign Material	2387
Other Microbial	1156
Labeling / Packaging Error	1008
Process / Temperature Control	896
Chemical Contamination	460
E. coli	456

```
In [81]: clean_data[["classification_days",
                    "recall_year"]].describe()
```

	classification_days	recall_year
count	29469.000000	29471.000000
mean	90.799518	2017.904007
std	3831.548106	11.307572
min	0.000000	212.000000
25%	26.000000	2015.000000
50%	38.000000	2017.000000
75%	64.000000	2022.000000
max	657475.000000	2026.000000

Validation caught one record dated year 212: a 657,475-day lag

Exploratory Data Analysis

29,471
recalled products in 7,898 events

44.0%
Class I: serious harm possible

94.9%
of recalls already terminated

33.7%
classified within 30 days

Pandas
value_counts, groupby, crosstab

NumPy
np.where, NaN handling

Matplotlib
bar, barh, stacked bar

Causes: undeclared allergen 32.3% and Listeria 25.1% lead, then Salmonella 10.1% and foreign material 8.1%

Products: Dairy & Eggs 20.4%, Seafood 14.1%, Fresh Produce 12.2% (24.3% fall in "Other Food")

Severity: Class II 50.0%, Class I 44.0%, Class III 6.0%

Firms: Garden-Fresh Foods tops the raw list (633), but Newly Weds Foods reaches 657 once its two spellings are merged

States: California 13.7% (4,045); the top 10 states hold 59.7%

Process: 98.6% voluntary; 38.3% notified by two or more channels; median 38 days to classify

```
In [54]: firm_counts = clean_data["recalling_firm"].value_counts()
         firm_counts.head(10)
```

```
recalling_firm
Garden-Fresh Foods, Inc.              633
Newly Weds Foods Inc                  438
FRESH IDEATION FOOD GROUP, LLC ...    409
Good Herbs, Inc.                      353
C & S Wholesale Grocers, Inc.         330
Whole Foods Market                    313
Blue Bell Creameries, L.P.            294
Reser's Fine Foods, Inc.              283
Dole Fresh Vegetables Inc             230
Newly Weds Foods, Inc.                219
```

Top firms: two spellings of one company appear as separate rows

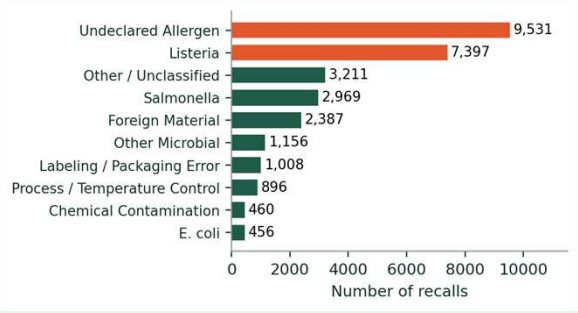
```
In [63]: print(clean_data["classification"].value_counts())
         print(clean_data["timely_classification"].value_counts())
```

```
classification
Class II              14739
Class I               12968
Class III              1762
Not Yet Classified        2

timely_classification
No                 19537
Yes                 9932
Not Available          2
```

Severity class and 30-day classification counts

Data Visualization

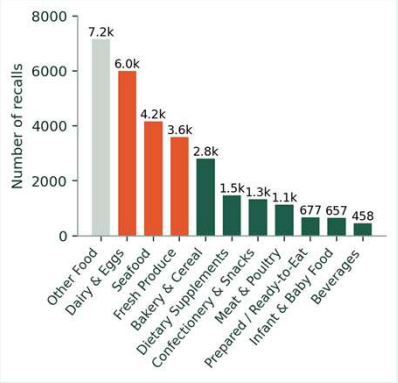


Recalls by Root Cause

Purpose: Rank what triggers recalls.

Shows: Allergens 9,531 and Listeria 7,397 lead the list.

Insight: Together they cause 57.4% of all recalls.

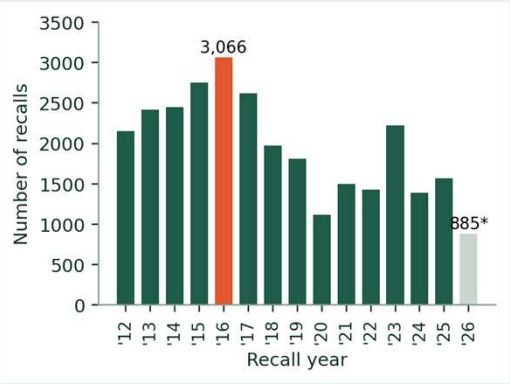


Recalls by Product Category

Purpose: Find where recalls concentrate.

Shows: Dairy & Eggs 6,005, Seafood 4,163, Fresh Produce 3,598.

Insight: Three categories hold 46.7% of recalls.

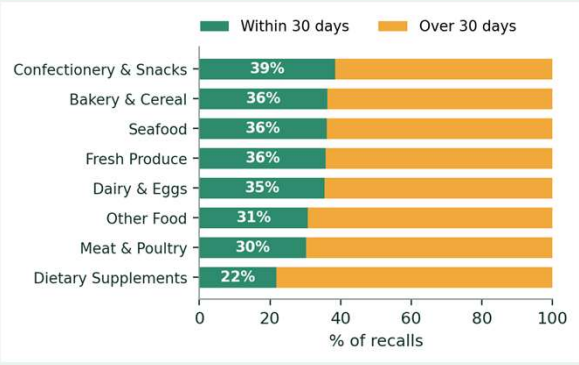


Recall Volume by Year

Purpose: Show the trend over time.

Shows: Peak 3,066 (2016), low 1,117 (2020), 2,221 in 2023.

Insight: No steady rise; 2026* covers Jan-Sep only.



Timely Classification by Product

Purpose: Compare how fast FDA classifies.

Shows: Share classified within 30 days, top 8 categories.

Insight: Only 22% for Dietary Supplements vs 39% for Confectionery.

Matplotlib charts from Food_Recalls_Cleaned.csv. * 2026 is a partial year. Years before 2012 (116 records) are left out of the year chart.

Insights, Recommendations & Conclusion

Key insights from the data

- Undeclared allergens (32.3%) and Listeria (25.1%) drive 57% of recalls
- Most Listeria (82%), Salmonella (85%) and E. coli (66%) recalls are Class I
- Dairy & Eggs, Seafood and Fresh Produce hold 47% of recalls
- Only 33.7% are classified within 30 days (median 38 days)
- Volume peaked in 2016 (3,066), dipped in 2020 (1,117), rebounded in 2023 (2,221)
- 24% of products and 11% of reasons stay "Other"; firm names have spelling variants

Recommendations

- Tighten allergen label checks and line changeover controls
- Strengthen Listeria and Salmonella monitoring and supplier checks
- Prioritize audits of dairy, seafood and fresh-produce suppliers
- Standardize recall data capture: fixed reason lists and firm IDs

What we achieved

Audited and cleaned 29,471 FDA recall records without dropping any, built 5 new features and 11 charts, and exported a validated 29-column CSV.

How it helps

Shows QA teams where recalls concentrate (allergens, Listeria, dairy, seafood, produce) so controls target the biggest risks first.

Limits & future scope

Keyword groups and the 30-day limit are approximations; FDA-reported recalls only. Next: parse quantities, merge firm names, add statistical tests and a Nestlé benchmark.